

Mobile Communication

Summer semester 2026

Exercise sheet 02

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- Release date: **2026-05-08**
- Submission of solutions: **No later than 2026-05-17 at 23:59.**
 - Submission is done online via your submission group's GitLab project. Submissions via other media (e.g. e-mail) are not accepted.
 - Flavius creates a Merge Request with a corresponding branch for every sheet. You do not need to merge the MR, just **push your solution onto the corresponding branch**. Please research further documentation if these concepts are not known yet.
 - **Every solution** must include a **single PDF file** containing all your documentation / explanations along with the names of all group members.
 - ... must include **all scripts/source files** (if any) used for solving the programming exercises.
 - ... as well as **sufficient documentation** on how to (compile and) run your code (if any), including all necessary packages (if applicable, please keep it sensible) and commands (**with all parameters**) for properly demonstrating your program. We require you to submit this in the form of a Makefile. A tutorial may be found here: makefiletutorial.com
 - In order to get admitted to the exam, you need to hand in solutions and achieve at least 50% of the total points. Additionally, according to the official admission rules you need to get at least 20% of points on 70% of sheets, which basically translates to what we stated in the orga slides: "We encourage you not to skip the latter sheets"
- Presentation of solutions: **During the tutorials on 2026-05-19**
- Next sheet will most likely be released on 2026-05-19

Exercise 1: Basics of IEEE 802.11 medium access [30 Points]

Section 3.1 of the MoCo lecture introduces the IEEE 802.11 Wireless LAN protocol. In contrast to the CSMA/CD medium access control method used in wired Ethernet, WLAN uses *Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)*. Specifically, CSMA/CA is realized by using the *Distributed Coordination Function (DCF)*.

- a) Briefly explain the working principle of the DCF. How is this method different to CSMA/CD in Ethernet and which problems are solved by this approach?

- b) The MoCo lecture mentions *Interframe spaces* and *Backoff time*. What is the duration of a *SIFS* and *DIFS* in the IEEE 802.11b standard? Give a precise definition of how the backoff time is calculated.
- c) What medium access mechanism can you think of that ensures that multiple stations and end devices can use the same medium together with allotted slots? How can “full-duplex communication” be achieved in this scenario?

Exercise 2: Wireless LAN and CSMA/CA [20 Points]

For this task you can assume that all WLAN stations are in communication range to each other. This means that we **do not** have the situation of hidden stations (aka hidden terminal).

- a) Explain in few words/sentences why it is not possible for two or more WLAN stations to detect a collision while the stations are sending themselves.

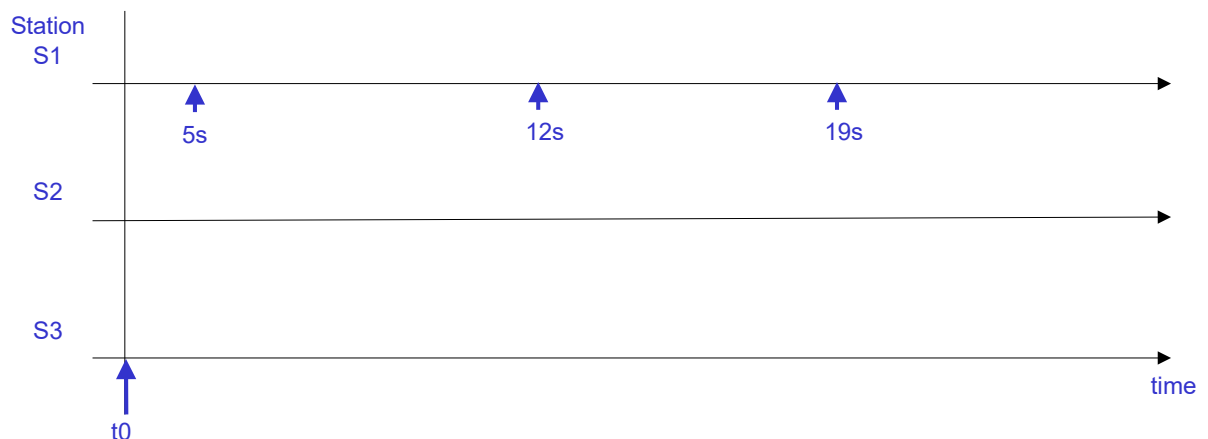
For the following subtasks, you need to complement the given timing diagrams and indicate:

- when a station is waiting for an inter frame space (SIFS or DIFS),
- when a station is waiting for a random backoff-time,
- and when a station is sending a data frame (DATA) or an acknowledgement frame (ACK)

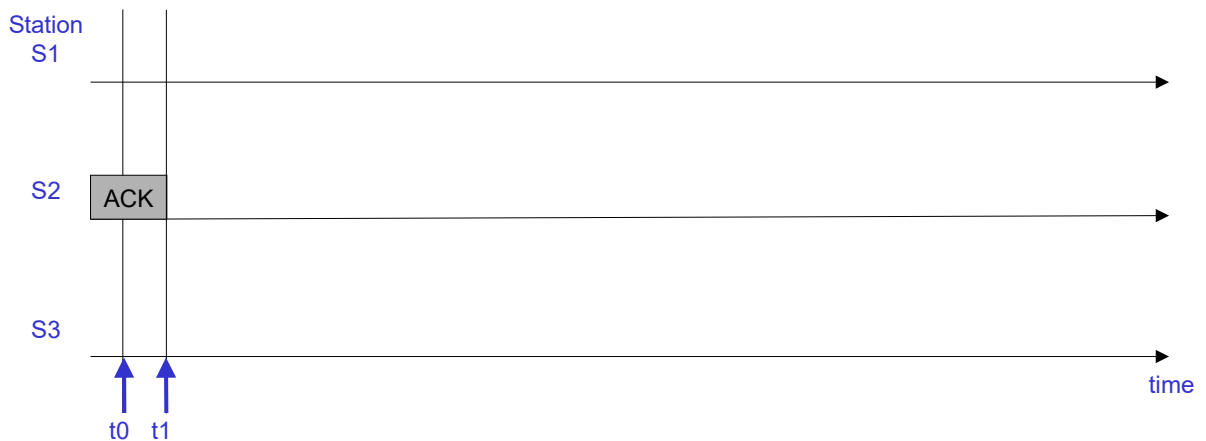
Your drawing does not need to be true to scale, but it should be clearly readable what is happening at which point in time. The resulting diagrams should look similar to the WLAN timing diagrams known from the lecture slides.

The random backoff-time is chosen as a (integer) multiple of a given (fixed) slot time (of some microseconds). Indicate this integer value when you use the chosen random, elapsed or remaining backoff time in your solutions.

- b) Look at the following given timing diagram. The medium is idle before t_0 and during the first 5 seconds. Only station S1 wants to send one DATA frame each at the three points in time indicated by the small arrows, at 5 seconds, at 12 seconds and at 19 seconds. The destination of those DATA frames is station S2. The transmission of a DATA frame only takes a few tens of milliseconds. And again, your drawing does not need to be true to scale. Complement the diagram for the given situation, indicating all times/intervals of waiting and sending, including DATA and ACK.



- c) Now look at the next given timing diagram. The medium is busy at t_0 because station S2 is sending an ACK frame (referring to a previous successful DATA frame transmission). The transmission of this ACK ends at t_1 . All three stations S1, S2 and S3 have each exactly one DATA frame ready for transmission. These DATA frames got ready for transmission between t_0 and t_1 . Complement the diagram starting at t_1 to result in a situation that first of all one of the three stations successfully transmits its DATA frame, and then the two other stations get into a collision! You can choose the receiver of each DATA frame from the three stations. Indicate when the collision takes place and which stations are involved. You should end your diagram extension with the last inter frame space after the collision.



Exercise 3: Simulation and analysis of wireless network medium access [50 Points]

A network hardware manufacturer has developed a new protocol for cheap wireless networking devices. In contrast to the CSMA/CA and DCF mechanisms used in Wireless LAN, the medium access mechanism of the new protocol is rather simple:

The medium is divided into time slots, which are determined by a coordinator node. Each time slot spans $100\mu s$ and starts with a $10\mu s$ synchronization frame sent by the coordinator node.

After the synchronization frame, each other node may try to send a data frame with a maximum length of $80\mu s$. Each data frame contains:

- MAC address of the source (6 byte)
- MAC address of the destination (6 byte)
- Data payload (0-84 byte)
- CRC checksum (4 byte)

If the data frame is received correctly by the destination node, it will use the last $10\mu s$ of the slot to send an acknowledgement frame to the source node.

- a) Which *channel data rate* must be used in order to transmit the complete data frame with maximum payload size in the given time interval? We are also interested in the *user data rate*. Give clear formulas and explain them.

How much overhead does this communication scheme have (i.e. how is the relation between channel and user data rate)?

What happens if we keep the same channel data rate, but increase or decrease the slot length? Explain your results!

- b) Obviously, if several nodes try to send data in the same time slot, there will be a collision. Collisions are detected by a missing acknowledgement at the end of the time slot. If a collision occurs, the nodes must wait a *random backoff interval*. This interval is defined as $[0, 2^i - 1]$ time slots, where i is the number of transmission failures for each data frame. After three unsuccessful retransmissions (i.e. when $i \geq 3$) the frame will be dropped and the node will attempt to send the next frame in its transmission queue.

Implement a *simulation for the communication of multiple nodes* which adheres to the protocol as defined above. The simulation should output a detailed log about what happens in each time slot, including information about retransmissions, queue length, etc. Furthermore, the simulation should accept the following input parameters:

- simulation duration t (number of time slots)
- number of nodes N
- queue probabilities $p_1, \dots, p_n$ which defines for each node $n \in N$ the likelihood that a new data frame is queued in a time slot.

Test your simulation with the following parameters and interpret the results:

- $t = 200, N = 3, p_1 = p_2 = p_3 = 0.1$
- $t = 200, N = 3, p_1 = p_2 = p_3 = 0.3$
- $t = 200, N = 3, p_1 = p_2 = p_3 = 0.5$
- $t = 200, N = 3, p_1 = 1, p_2 = p_3 = 0$

- c) Finally, we want to have a closer look at the efficiency of this protocol. Using your existing simulation, create a series of XY plots which show the achieved user data throughput (Y-axis) depending on the queue probability (X-axis).

Use the following parameters:

- $t = 1000, N = 2$
- $t = 1000, N = 5$
- $t = 1000, N = 10$

Interpret the resulting plots and comment on the performance of this protocol. Give an estimation how the CSMA/CA mechanism of Wireless LAN would perform in similar situations!